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Emotion Classification Project

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Introduction

- Natural Language Processing
- Key Challenge
- Use Case

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Problem statment
And Client Value

- Content Intelligence Agency
- Goal
- Value

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Data Overview

Data sources:

- GoEmotion
- Friends
- Twitter
- All Groups

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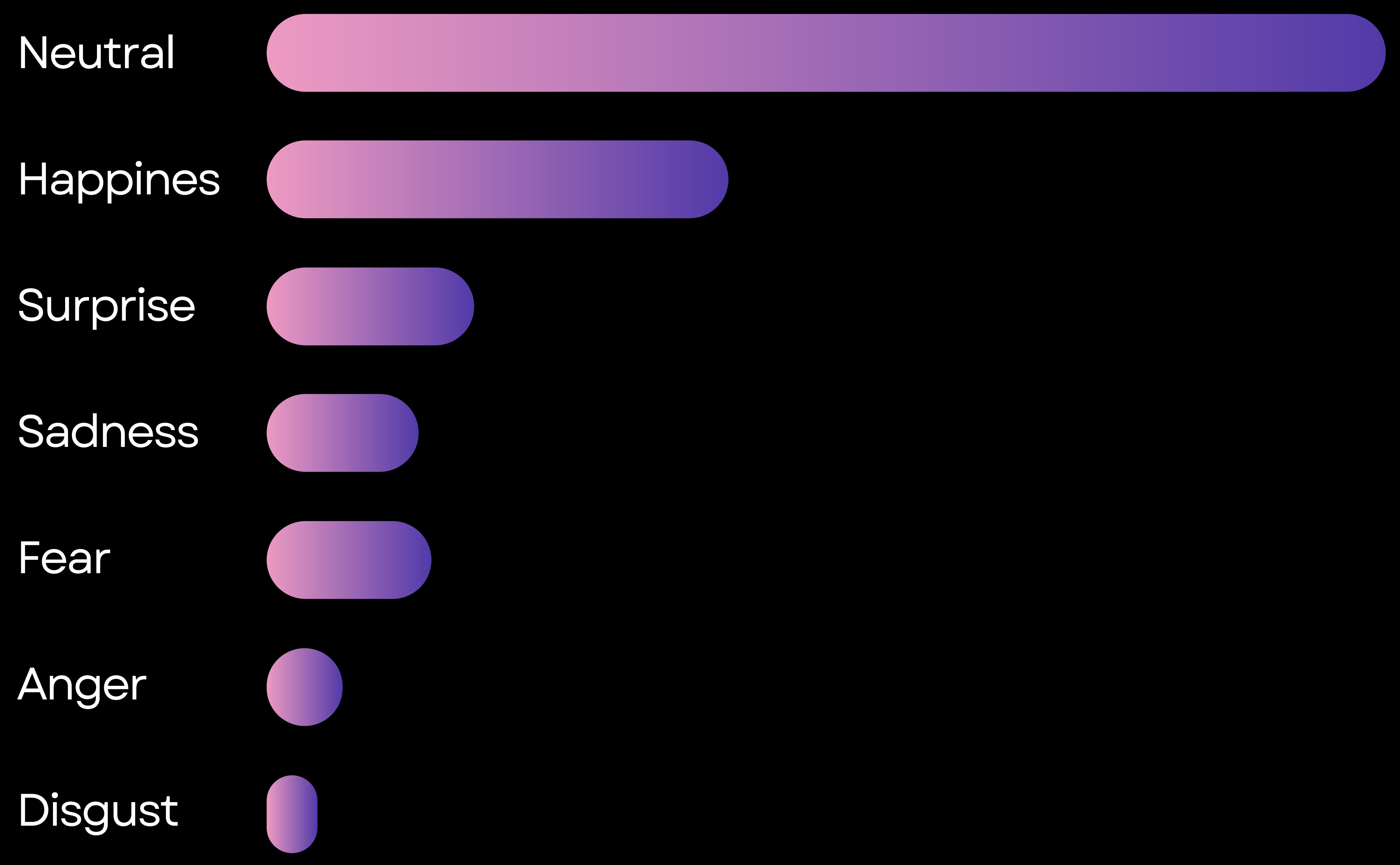
Data Overview



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Data Overview

Pre-processing steps	Raw	Light	Medium	Heavy
Lower casing		x	x	x
Fixing contractions		x	x	x
Stripping extra space		x	x	x
Remove special characters			x	x
Removing stop words				x
Steming				x
Lemmatization				x

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Data Overview

Features extraction steps:

- POS tags extraction with NLTK
- TF-IDF calculation
- Sentiment Analysis (TextBlob & VADER)
- Pre-trained word embedding using GloVe 300 dimentions

- Readability - Flesch score
- Complexity metrics
- Extract named entities
- Sentence and words length
- Exclamation marks count

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Data Limitations

- Missing consistent data
- Wrong labelling
- Only signs or one word sentences
- Imbalance

MODEL

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Model(s) - Vik

- Algorithms chosen and why
- Any alternatives considered
- Libraries / tools

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LSTM

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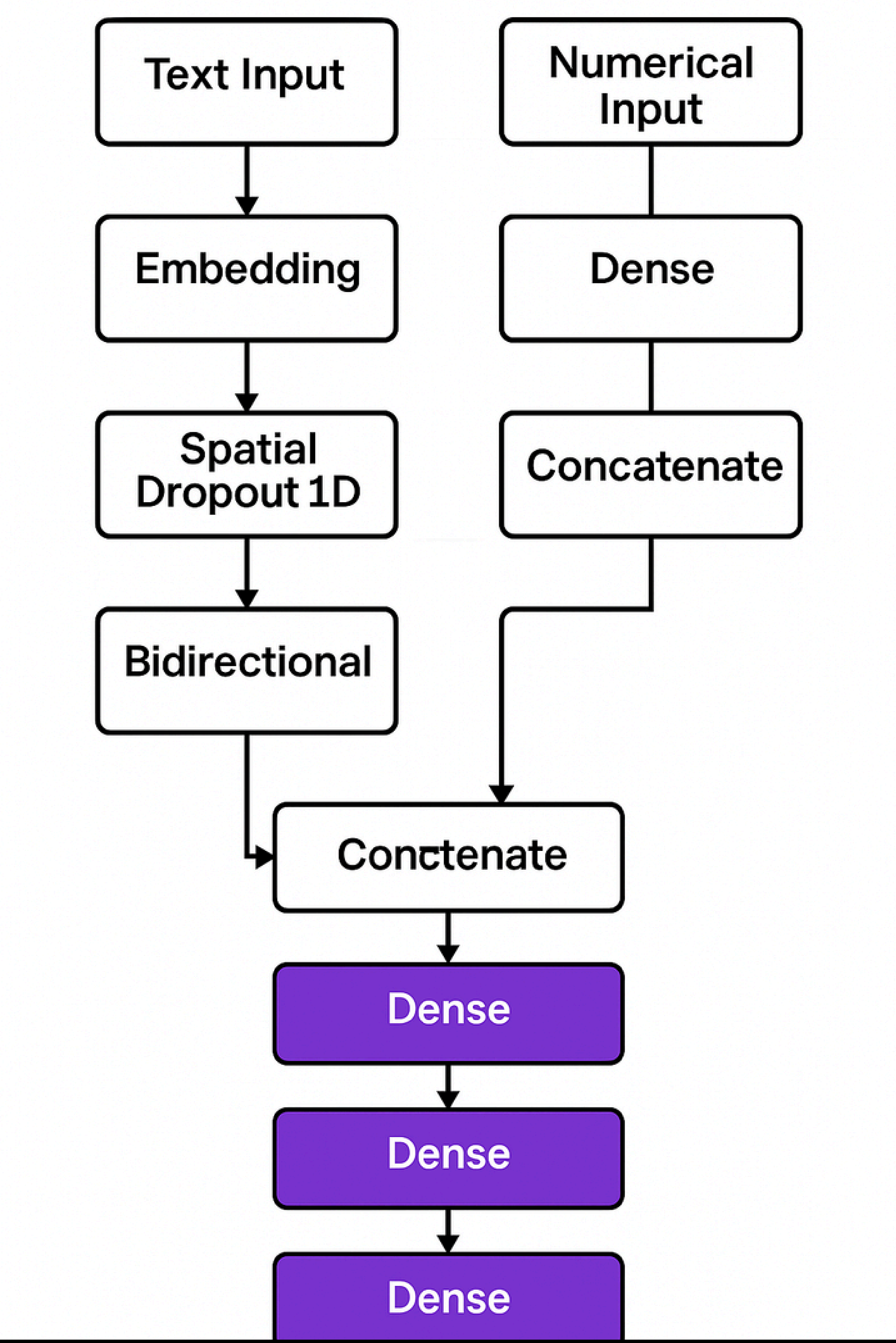
LSTM

- All groups data 24,000 rows
- NLTK tokenizer
- Embedding - GloVe 300

- Text
- Vader: neg, neu, pos, compound, intensity
- textblob polarity, subjectivity, intensity



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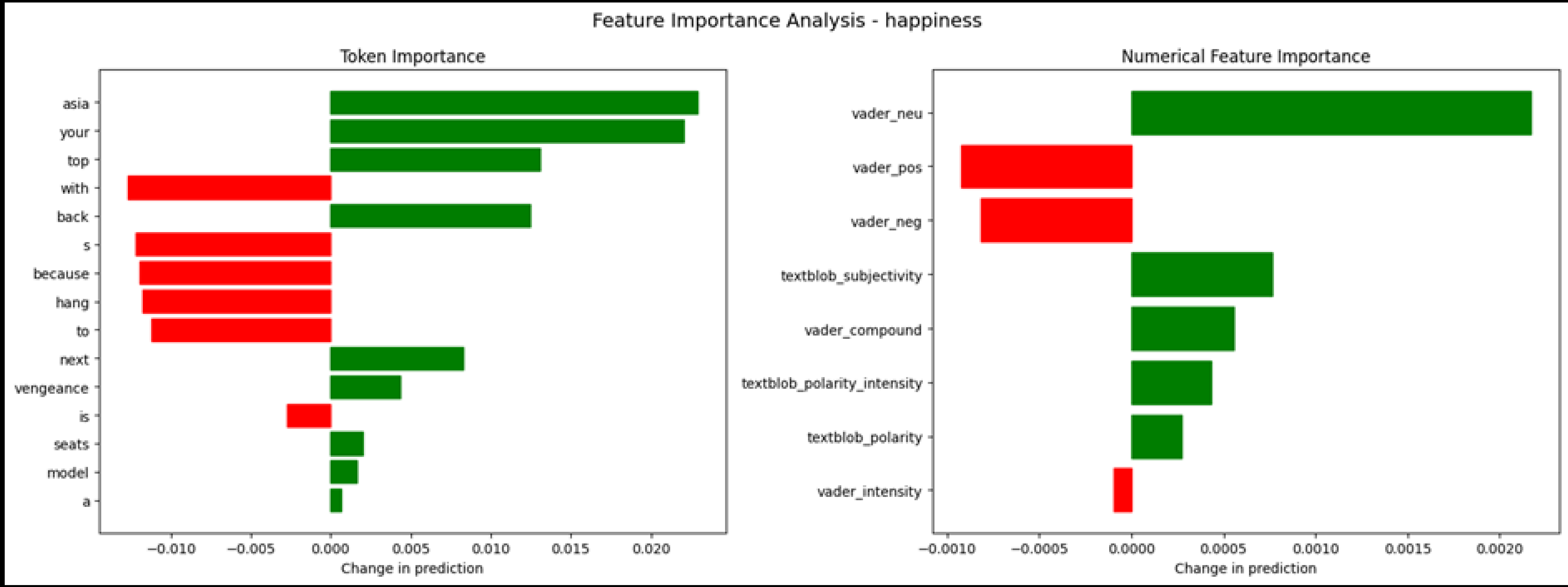
LSTM



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i just **hope** that there s **less drama**
True emotion: neutral
Predicted emotion: happiness

Text: weird model **but** that
True emotion: neutral
Predicted emotion: **surprise**

Text: when i look around all of the girls the girl who has the **best body** is **me**
True emotion: neutral
Predicted emotion: **happiness**

Text: like do this do that
True emotion: neutral
Predicted emotion: surprise

Text: and **who** s **moving out**
True emotion: neutral
Predicted emotion: surprise

Text: **what second**
True emotion: neutral
Predicted emotion: surprise

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Class	Precision	Recall	F1-Score	Support
happiness	0.128	0.615	0.212	39
Sadness	0.000	0.000	0.000	2
Anger	0.161	1.000	278	5
Fear	1.000	0.429	0.600	7
Surprise	0.066	0.750	0.121	8
Neutral	0.982	0.666	0.793	799
Disgust	0.000	0.000	0.000	0
Accuracy			**0.663**	860
Macro Avg	0.390	0.577	0.334	860
Weighted Avg	0.927	0.663	0.754	860

MODEL

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Model(s) Used - Soheil

- Algorithms chosen and why
- Any alternatives considered
- Libraries / tools

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Model Evaluation Metrics

- Explain selected metrics (accuracy, precision, recall, F1-score)
- Trade-offs between metrics
- What was prioritized and why

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Model Results

- Graphs / tables of results
- Compare models if applicable
- Interpretation of the numbers

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Model Results (Visuals)

- Graphs / tables of results
- Compare models if applicable
- Interpretation of the numbers

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XAI

- Balance between performance & complexity (XAI)

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Pipeline Overview

- Show your pipeline flow diagram
- Data collection → preprocessing → modeling → evaluation

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Strengths & Weaknesses

- What worked well
- can be deployed on edge devices
- fast, accuracy vs consumption - cost reduction, minority classes classification need improvement
- What challenges were faced
-

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Ethical Considerations

- Data privacy, bias in data, fairness
- Relation to chosen metrics and model behavior

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Implementation Limitations

- Scalability concerns
- Real-world constraints (data availability, costs, etc.)

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Next Steps & Recommendations

- Improvements for future
- Recommendations for the client
- Possible extensions of the project

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QUESTIONS?

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Thank You